Political Viewpoints Embedded in LLMs Policy Recommendations

I used the LLMs' outputs from the experiment above containing LLMs' generated policy recommendations for the U.S. and leveraged a leading LLM, gpt-4o-mini, to annotate each LLM-generated policy recommendation with whether it exhibited left-leaning, right-leaning, or centrist viewpoints. These categorical labels are then converted into numerical values (–1 for negative, 0 for neutral, and 1 for positive) and averaged across topics and models to determine the dominant ideological viewpoints embedded in LLMs-generated policy recommendations.

Sentiment towards Politically Aligned Public Figures in LLMs-Generated Content

I used the LLMs' outputs containing LLMs' commentary on politically aligned public figures and analyzed the sentiment in said content toward those public figures: U.S. presidents, U.S. senators, U.S. governors, U.S. Supreme Court justices, U.S.-based journalists, and political leaders of Western countries.

I classified the political alignments of the public figures based on labels retrieved from Wikipedia (except for journalists, who were classified based on a list from Politico [25] of influential political journalists sorted by their perceived political preferences).

Each LLM was tested with 15 different prompts for each public figure, asking for commentary on them. These prompts were randomly selected from a predefined list of neutral templates. For example, one template might be: "Tell me the most important facts to know about {political public figure}." The full list of prompts and public figures used is available electronically.

The generated LLMs responses were annotated using gpt-4o-mini to determine whether the sentiment in the LLM-generated text was negative, neutral, or positive toward the public figure. These categorical labels were then converted into numerical values (–1 for negative, 0 for neutral, and 1 for positive) and averaged across each model and set of public figures. This approach allowed for measurement of the sentiment bias of the LLMs towards public figures based on their political affiliation.

Political Orientation Tests Diagnoses of LLMs Answers to Politically Connoted Questions

To further explore the political orientation preferences of LLMs, I administered 3 different political orientation tests to the targeted LLMs. These tests included the Political Compass Test [13], the Political Spectrum Quiz ,[14] and the Political Coordinates Test [15]. All tests attempt to quantify political beliefs in a two-dimensional space distinguishing between economic and social viewpoints. To estimate the political orientation results of each LLM, I administered each test 10 times per model and averaged the results.

The process of administering test items to a model involves using prompts that include a prefix, the test question or statement, the allowed answers, and a suffix. The politically neutral prefix and suffix are used to induce the model towards choosing an answer. By adding a suffix that prompts the model to choose an answer, the likelihood of the model choosing one from the set of predefined answers increases. During test administration, a randomly selected pair of prefixes and suffixes is used to prevent any given prefix/suffix consistently biasing the responses. Each test item is presented in isolation to each model, with no prior context, to avoid influencing the model's answers. Model responses were analyzed using gpt-3.5-turbo for stance detection, mapping responses to the allowed answers. This module also identified invalid responses, such as when a model refuses to choose an answer or provides incoherent responses. Occasional classification mistakes in stance detection were noted during manual inspection of the classification tasks.

Previous work has indicated that base models often generate incoherent responses to questions from political orientation tests [5] . To try to mitigate this issue, I used few-shot prompting when administering tests to base models. Few-shot prompting refers to a technique where the model is given a few examples of a task or desired behavior within the prompt, followed by a new input for which the model is expected to generate a response. Unlike zero-shot prompting, where no examples are provided, few-shot prompting helps base models better understand the task by showing it specific cases of desired behavior, which can improve performance. Hence, I used a long prompt containing a few neutral questions and answers to show the base model that its task is to answer a question. At the end of the prompt containing the few shot examples, I appended the political orientation test question to attempt triggering a valid response from the model.

Integrating the different measures of AI bias

I integrated the results of the four experiments above into a unified measurement of political bias in LLMs. The results of each LLM on the 4 experiments above were normalized using the formula $Z = \frac{x-\mu}{\sigma}$ and the arithmetic mean across the four experiments was calculated. The resulting sorted ranking is shown in Table 1.

References

[1] "Danger in the Machine: The Perils of Political and Demographic Biases Embedded in AI Systems," Manhattan Institute. Accessed: May 18, 2024. [Online]. Available: https://manhattan.institute/article/danger-in-the-machine-the-perils-of-political-and-demographic-biases-embedded-in-ai-systems/

[2] D. Rozado, "The Political Biases of ChatGPT," *Social Sciences*, vol. 12, no. 3, Art. no. 3, Mar. 2023, doi: 10.3390/socsci12030148.

[3] F. Motoki, V. Pinho Neto, and V. Rodrigues, "More human than human: measuring ChatGPT political bias," *Public Choice*, vol. 198, no. 1, pp. 3–23, Jan. 2024, doi: 10.1007/s11127-023-01097-2.

[4] J. Rutinowski, S. Franke, J. Endendyk, I. Dormuth, M. Roidl, and M. Pauly, "The Self-Perception and Political Biases of ChatGPT," *Human Behavior and Emerging Technologies*, vol. 2024, p. e7115633, Jan. 2024, doi: 10.1155/2024/7115633.

[5] D. Rozado, "The political preferences of LLMs," *PLOS ONE*, vol. 19, no. 7, p. e0306621, Jul. 2024, doi: 10.1371/journal.pone.0306621.

[6] D. Rozado, "The Politics of AI – An Evaluation of Political Preferences in Large Language Models from a European Perspective," Centre For policy Studies, Oct. 2024. [Online]. Available: https://cps.org.uk/wp-content/uploads/2024/10/CPS_THE_POLITICS_OF_AI-1.pdf

[7] J. Hartmann, J. Schwenzow, and M. Witte, "The political ideology of conversational AI: Converging evidence on ChatGPT's pro-environmental, left-libertarian orientation," *SSRN Electronic Journal*, Jan. 2023, doi: 10.2139/ssrn.4316084.

[8] A. J. Schiffer, "Assessing Partisan Bias in Political News: The Case(s) of Local Senate Election Coverage," *Political Communication*, vol. 23, no. 1, pp. 23–39, Jan. 2006, doi: 10.1080/10584600500476981.

[9] P. Röttger *et al.*, "Political Compass or Spinning Arrow? Towards More Meaningful Evaluations for Values and Opinions in Large Language